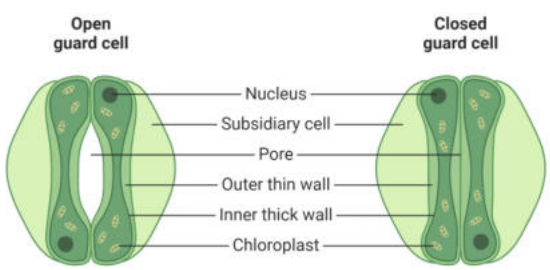
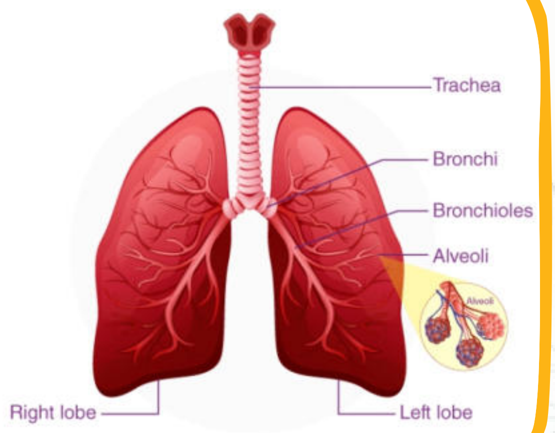
UNIT NO. 1

GASEOUS EXCHANGE





GASEOUS EXCHANGE IN PLANTS

Plants, and animals, engage in critical processes that necessitate gaseous exchange. Despite lacking respiratory organs like humans, plants have evolved intricate mechanisms for the exchange of gases to sustain life. Two fundamental processes in plants, photosynthesis, and respiration, serve as primary drivers of this exchange. We shall explore and contrast these processes from a more professional standpoint.

COMPARISON:

ASPECT	PHOTOSYNTHESIS	Respiration
Function	Photosynthesis is akin to a culinary artist crafting sustenance for the plant.	Respiration operates as an internal powerhouse, generating energy.
Substrates	Photosynthesis utilizes carbon dioxide and harnesses sunlight as its core ingredients.	Respiration relies on oxygen and organic compounds, typically glucose.
Light Dependency	Photosynthesis is light-dependent, requiring the presence of sunlight.	Respiration, in contrast, occurs independently of light, functioning in both day and night.
Location	Photosynthesis primarily transpires within chloroplasts of plant cells, particularly in leaves.	Respiration is a ubiquitous process, transpiring in all living plant cells.
Gas Exchange	Photosynthesis assimilates carbon dioxide (CO_2) and liberates oxygen (O_2) as a byproduct.	Respiration involves oxygen (O_2) intake and the release of carbon dioxide (CO_2) as waste.
Temporal Pattern	Photosynthesis predominantly occurs during daylight hours.	Respiration, in contrast, operates continuously, devoid of diurnal variation.
Chlorophyll Role	Photosynthesis relies on chlorophyll, a pigment, to capture and convert sunlight into energy.	Respiration functions independently of chlorophyll, making it a universal process.

In summary, photosynthesis acts as a culinary artisan, harnessing carbon dioxide, sunlight, and chlorophyll to create sustenance, while respiration serves as a tireless, energy-generating powerhouse that operates continuously in all plant cells. Together, these processes sustain the vitality of the plant kingdom.

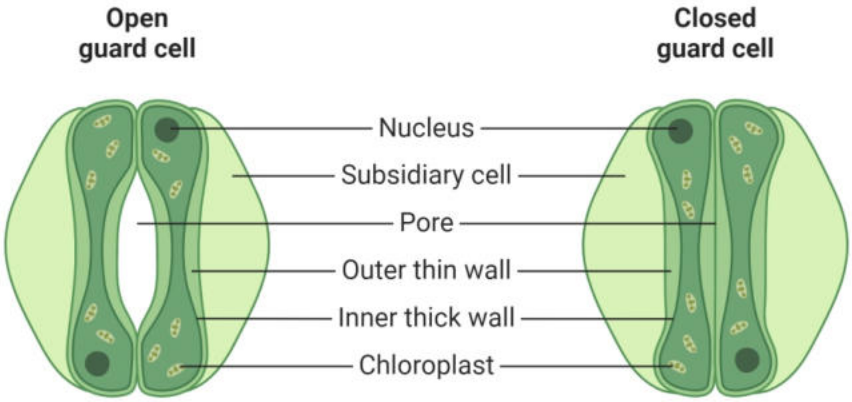
STOMATA:

What Are They?

• Tiny openings in leaf epidermis.
• Used for gas exchange in plants.

Structure:
• Formed by two guard cells.
• Guard cells contain chlorophyll.
• Thick inner and elastic outer cell walls.
Function:
• Opening and closing controlled by guard cell turgidity.
• Open during the day due to increased turgidity from photosynthesis.
• Allows intake of carbon dioxide (CO_2) and release of oxygen (O_2).
• Closed at night when photosynthesis stops.

Stomata, or **"mouths"** of leaves, are microscopic openings controlled by guard cells, enabling plants to exchange gases with the environment. They open during the day, taking in carbon dioxide and releasing oxygen, thanks to increased guard cell turgidity from photosynthesis. At night, they close when photosynthesis ceases.



Practical Activity:
The effect of light on the net gaseous exchange from leaf by using Hydrogen bicarbonate as indicator.
Hydrogen bicarbonate is an indicator for carbon dioxide. Its colour turns as follows according to the level of carbon dioxide:

PHOTOSYNTHESIS	RESPIRATORY
HIGHEST	YELLOW
HIGH	ORANGE
ATMOSPHERIC LEVEL	RED
LOW	MARGENTA
LOWEST	PURPLE

Requirements:
Four test tubes, test tube stand, aluminum foils or black paper, tissue paper, fresh green leaves, four corks, wax, thread, glass marking pencil
Steps:
• Label four test tubes 1, 2, 3, and 4.
• Fill each test tube one-quarter full with hydrogen bicarbonate indicator.
• Suspend a leaf in each test tube, except test tube 3.
• Plug all the tubes with corks and seal them with wax.
• Wrap test tube 2 with aluminum foil or black paper, and wrap test tube 3 with tissue paper.

• Place all the tubes in a well-lit area.

Observations:

Test tube 1: The indicator will turn yellow, indicating the highest rate of photosynthesis and the lowest rate of respiration.

Test tube 2: The indicator will turn orange, indicating a high rate of photosynthesis and a low rate of respiration.

Test tube 3: The indicator will turn red, indicating the lowest rate of photosynthesis and the highest rate of respiration.

Test tube 4: The indicator will turn magenta, indicating a low rate of photosynthesis and a high rate of respiration.

CRITICAL THINKING:

Q1. Is there any change of coloration of Hydrogen bicarbonate indicator?

ANS: Yes, hydrogen bicarbonate indicator can change color depending on the concentration of carbon dioxide in a solution.

Q2. What account for these changes?

ANS: Hydrogen bicarbonate indicator changes color depending on the concentration of carbon dioxide. This can be used to measure the rate of photosynthesis and respiration in plants.

GASEOUS EXCHANGE IN ANIMALS

Animals, like plants, require gaseous exchange for respiration. They take in oxygen and release carbon dioxide to obtain energy from food. Aquatic animals exchange gases with water, containing about 5% oxygen, while terrestrial animals use air, which contains about 21% oxygen. Animals have specialized respiratory surfaces, which can be their body surface or internal structures, facilitating this vital exchange.

Respiratory Surfaces - Large Surface Area:

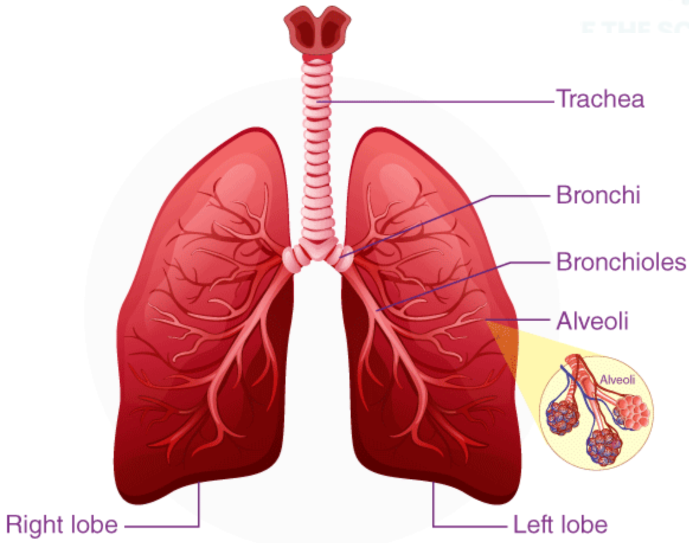
A larger surface area enables efficient gaseous exchange by promoting faster diffusion, which supplies oxygen and eliminates carbon dioxide. This compensates for the relatively small surface area-to-volume ratio found in many animal bodies.

GASEOUS EXCHANGE IN HUMANS

In humans, respiration encompasses breathing, gaseous exchange, and cellular respiration. Our respiratory surface, the alveoli, is located within paired lung organs inside the body.

Human Respiratory System:

Our respiratory system consists of paired lungs located inside the thoracic (chest) cavity and Air passage ways.



LUNGS:

Lungs, enclosed in pleural membranes, aid breathing with lubricating fluid. Protected by the ribcage and diaphragm, they house millions of alveoli, microscopic sacs for gas exchange with surrounding capillaries. The airway, starting from nostrils to bronchioles, is lined with ciliated cells and mucous for filtration. This ensures clean air reaches the delicate respiratory surfaces, enabling efficient oxygen exchange and carbon dioxide removal essential for human respiration.

TRACHEA:

The trachea, a long tube, connects to the nasal sacs and contains the vocal cords in the larynx. The glottis, its opening, is guarded by the epiglottis, preventing food entry during swallowing. C-shaped cartilaginous rings support the trachea, preventing collapse.

BRONCHI:

The trachea divides into two bronchi in the thorax, each with C-shaped cartilaginous rings. These bronchi further branch into smaller bronchioles within their respective lungs.

BRONCHIOLE:

Each bronchiole is very thin tube that opens into air sacs or alveoli.

Process of Breathing:

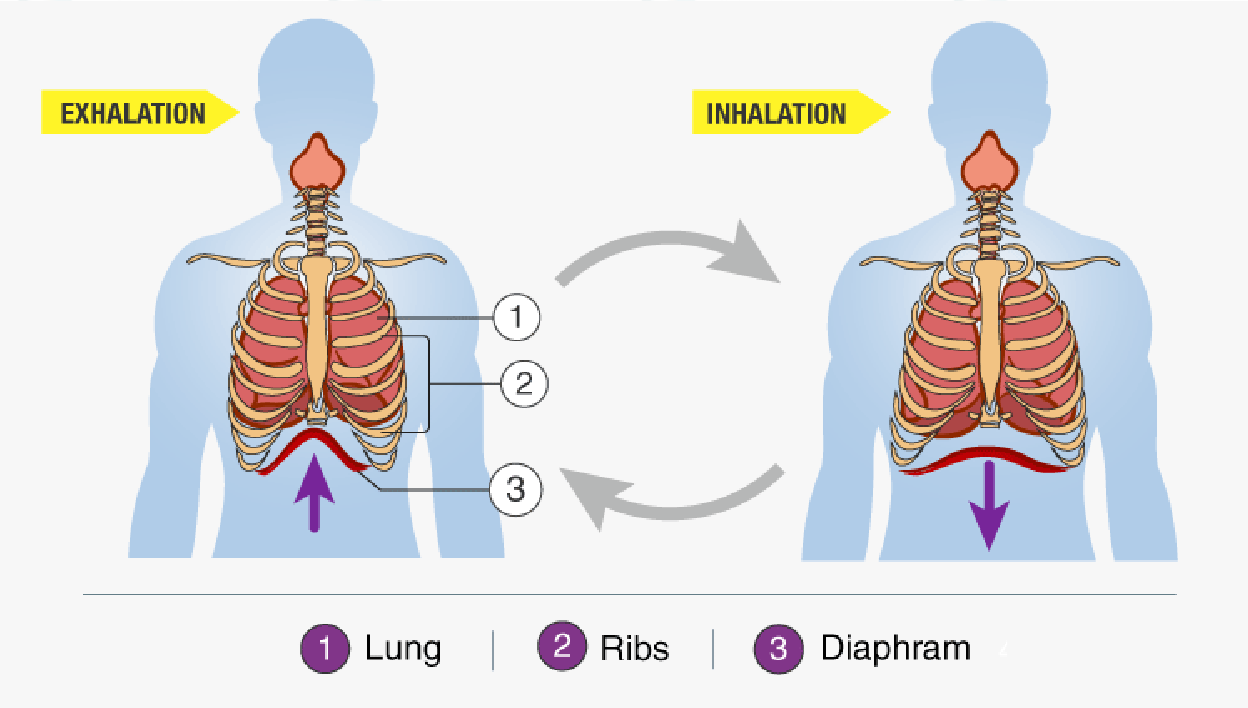
The exchange of gases within the lungs relies on the process of breathing or ventilation, as the respiratory surfaces are deep within the lungs. Breathing consists of two phases: inspiration, where air is drawn into the lungs, and expiration, where air is expelled from the lungs.

INSPIRATION:

During inspiration, the diaphragm and intercostal muscles contract, expanding the chest cavity and lowering lung pressure, allowing external air to rush in, expanding the lungs.

EXPIRATION:

In contrast to inspiration, expiration is the process of air moving out from the lungs. It occurs when both intercostal muscles and the diaphragm relax, causing the ribs to move inward and the diaphragm to flatten. This compresses the chest cavity, increasing pressure in the lungs, and forcing air out of the body.



GASEOUS EXCHANGE IN ALVEOLI

The alveoli are the primary sites for gaseous exchange within the lungs, where oxygen enters the bloodstream, and carbon dioxide is expelled. This intricate process enables the body to maintain the essential balance of gases crucial for cellular respiration and overall physiological functioning.

Composition of Inspired and Expired Air:

COMPONENT	INSPIRED AIR (%)	EXPIRED AIR (%)
Oxygen (O2)	21%	16%
Carbon Dioxide (Co2)	0.04%	04%
Nitrogen (N2)	78%	78%
Water Vapor (H2O)	Variable	Saturated (100%)
Temperature	Varies with environment	Body temperature (Approximately 37°C)

This table now includes the percentage of water vapor and mentions the temperature characteristics of both inspired and expired air.

Rate of Breathing at Rest and During Exercise:

Breathing is primarily an involuntary process, controlled by the hypothalamus in the brain. It adjusts automatically to changing internal or external conditions. During exercise, the rate of breathing increases to meet the heightened oxygen demand of the muscles, leading to a buildup of carbon dioxide in the blood, which further stimulates increased breathing. Prolonged exercise may shift to anaerobic respiration, generating lactic acid in muscles instead of carbon dioxide, causing discomfort. To repay the "oxygen debt," deep breaths supply the extra oxygen needed, aiding recovery.

ARTIFICIAL VENTILATOR:

An artificial ventilator is a machine that helps when a patient struggles to breathe naturally. It delivers oxygen-rich air directly into the trachea through a tube placed in the mouth, reaching the windpipe.

RESPIRATORY DISORDERS

These ailments impact the respiratory system, causing breathing difficulties and discomfort. Conditions such as bronchitis, emphysema, pneumonia, asthma, and lung cancer affect lung function, often arising from factors like smoking, pollution, or infections.

BRONCHITIS:

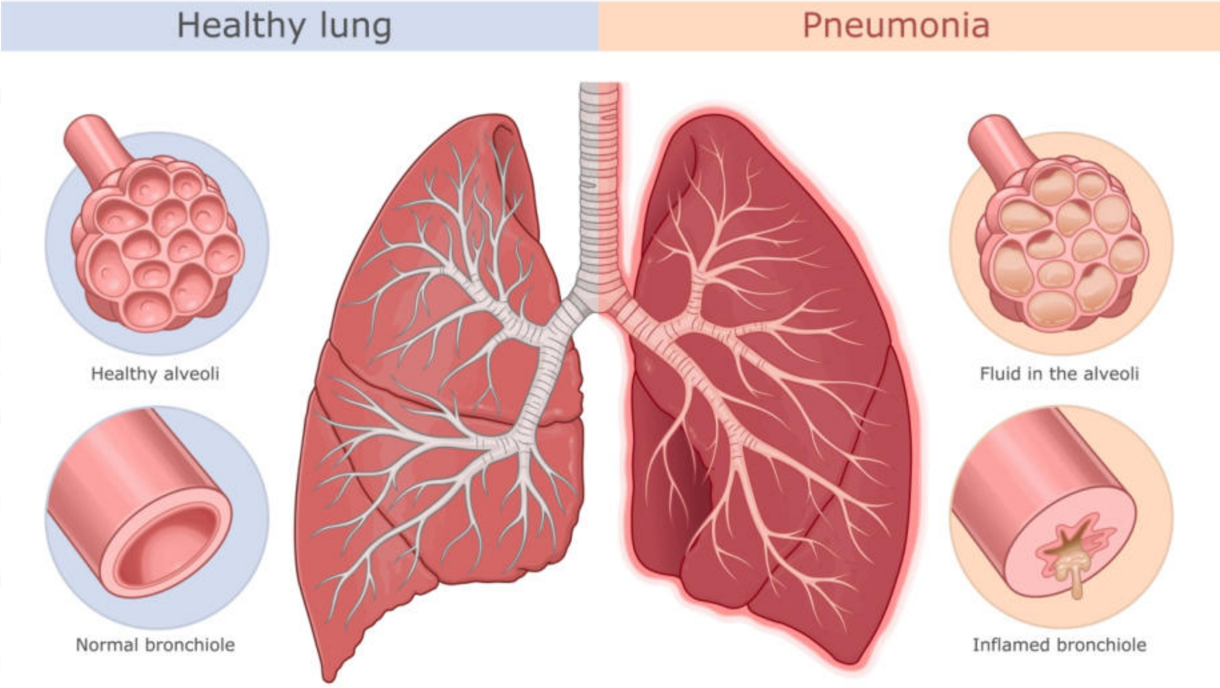
Inflammation of air passages, often due to smoking or bacteria, leading to cough, increased mucus, breathlessness, and low fever.

EMPHYSEMA:

Progressive destruction of alveoli from long-term exposure to industrial pollutants, causing difficult breathing and cough with phlegm.

PNEUMONIA:

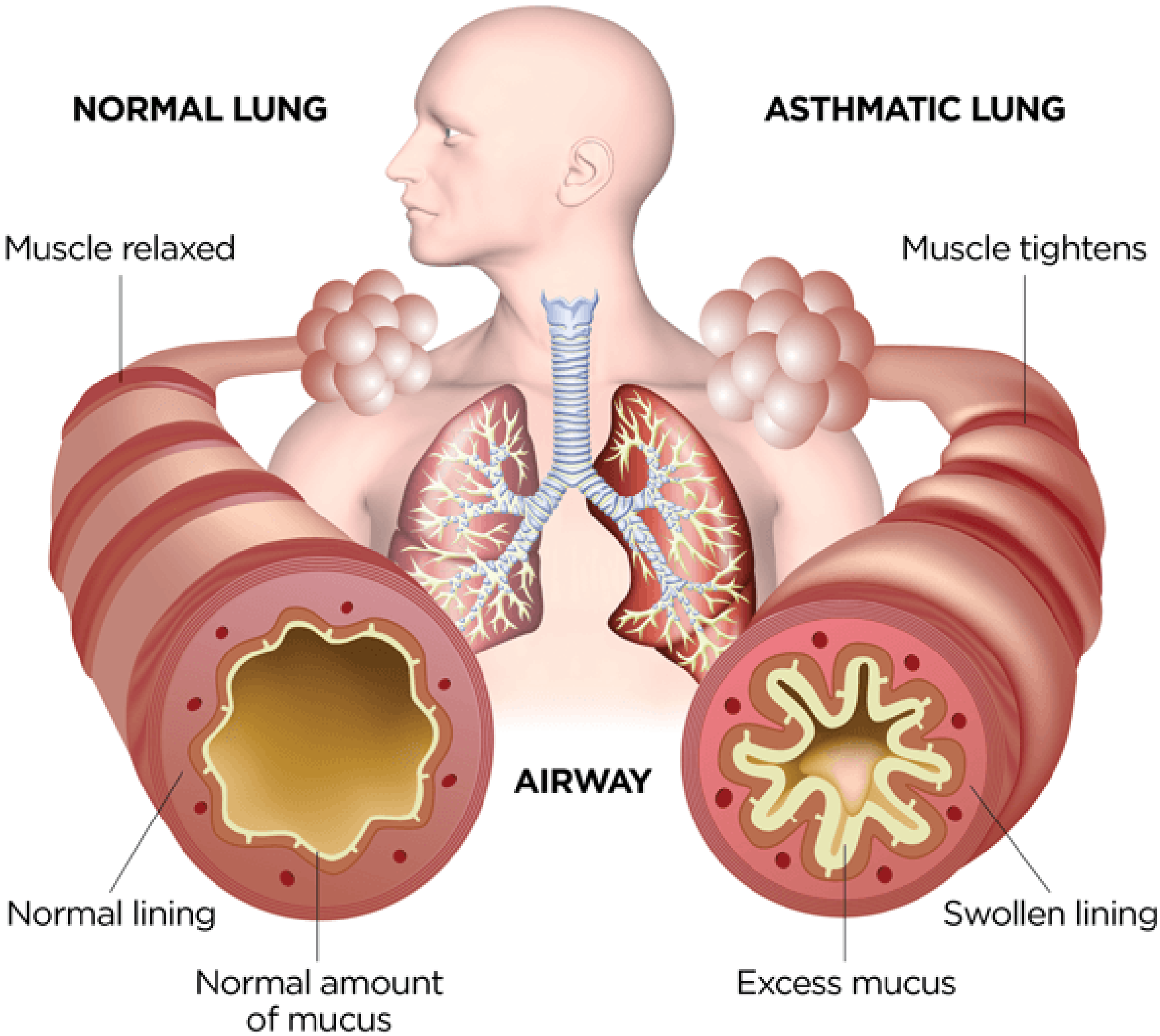
Infectious disease, usually from bacteria, viruses, or fungi, infecting alveoli, leading to fluid or pus buildup, difficulty breathing, fever, cough, chills, and chest pain.



ASTHMA:

Inflammatory condition of lung airways, causing shortness of breath, chest pain, fever, wheezing during exhalation, and cough. Often triggered by allergies to substances like pollen, dust, smoke, or fur, leading to airway obstruction.

LUNG CANCER:



Typically linked to smoking or air pollution, where abnormal cells in the lungs can spread to other tissues. Symptoms include bloody cough, breathlessness, recurring lung infections, weight loss, bone pain, hoarseness, weakness, and fatigue.